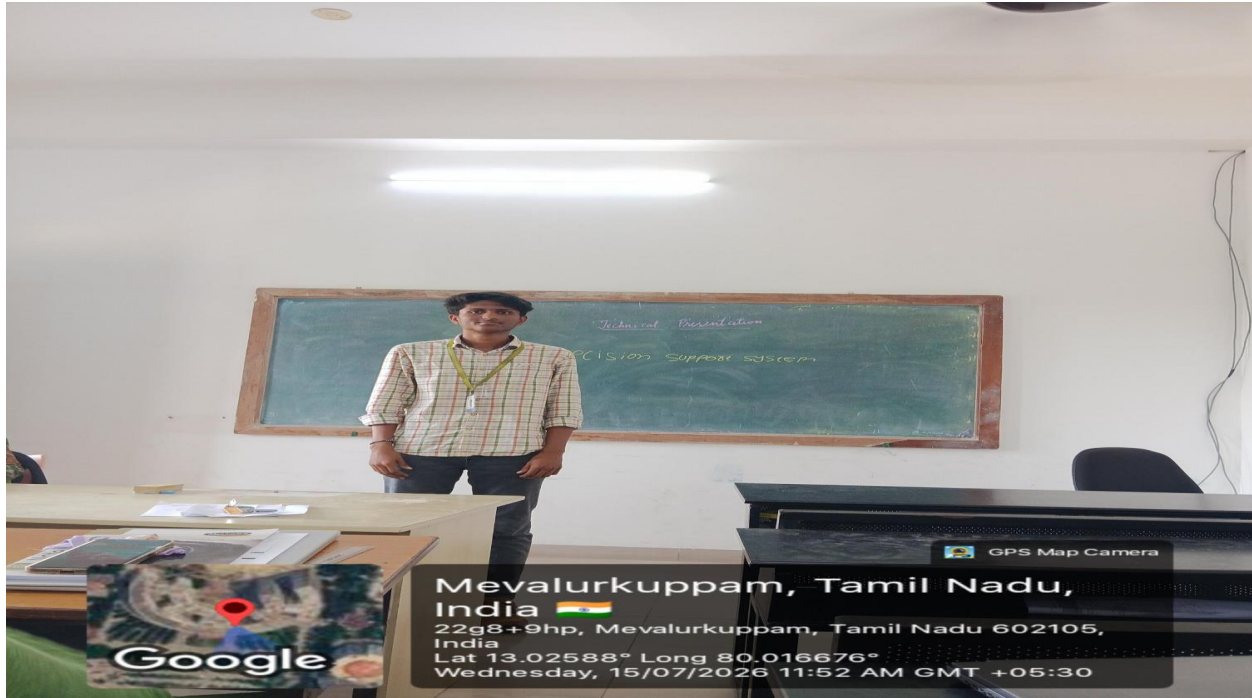


TECHNICAL PRESENTATION

OLAP and Data Cubes



Online Analytical Processing (OLAP) is a computer computing method that enables users to easily and selectively extract and query data in order to analyze it from different points of view. Unlike Online Transactional Processing (OLTP) which manages real-time day-to-day transactions, OLAP is optimized for complex analytical queries on massive amounts of historical data. It provides a consolidated, single source of truth that powers business intelligence (BI) and strategic decision-making.

Data Cube Structure

An OLAP Data Cube is a multidimensional array structure designed for high-speed data analysis. While a traditional spreadsheet represents data in two dimensions (rows and columns), a data cube extends this to three or more dimensions (hypercubes).

OLAP Operations

The four primary OLAP operations—Slice, Dice, Roll-up, and Drill-down—allow analysts to navigate multidimensional data cubes seamlessly to extract business insights.

Slice cuts a single, specific slice from the data cube by selecting a single value for one of its dimensions. For example, if a cube contains sales data across time, product, and geography, slicing for “2026” isolates a two-dimensional spreadsheet showing product sales across different geographic regions for that specific year only.

Dice extracts a smaller sub-cube from the main data cube by applying filters to multiple dimensions simultaneously. Instead of looking at all products and all regions, an analyst might dice the cube to view only laptops and smartphones, strictly within Mumbai and Delhi, during the first quarter.

Roll-up performs data aggregation by climbing up a dimension’s structural hierarchy or by removing a dimension entirely. It condenses highly detailed data into a broader summary, such as combining daily retail transactions to view total monthly sales, or aggregating city-level performance into a single country-wide metric.

Drill-down is the exact reverse of a roll-up, allowing analysts to step down a dimension hierarchy to uncover finer, more granular details. This operation breaks down broad summary data into smaller parts, such as clicking on total annual revenue to reveal the specific quarterly, monthly, or individual store-level transactions that generated it.

Applications in Analytics

Because OLAP pre-aggregates information, analysts can run complex queries instantly without waiting for traditional database joins. Common analytical applications include:

Financial Planning & Budgeting: Analysing corporate expense trends across different business departments and multi-year time horizons.

Sales & Marketing Forecasting: Evaluating product category performance across geographic regions to optimize localized supply chains.

Supply Chain Optimization: Tracking inventory levels, supplier turnaround timelines, and defect rates over time to limit overhead costs.